AGRINEGOTIATOR — PHASE 3, 4 & 5 VALIDATION REPORT

Buyer Demand, Semantic Matching, LangGraph Multi-Agent Negotiation & E2E Checkpoint

Phase 3 (Buyer Demand): PASS (7/7 Tested)	**Phase 4 (Matching Engine): PASS (3/3 Tested)**
Phase 5 (LangGraph AI): PASS (5/5 Tested)	**Phase 3→5 Integration: PASS (All 6 Steps)**
Phase 1–5 Defects: 0 Application Bugs	**Checkpoint Decision: READY FOR MANUAL TESTING**

1. PHASE 3: BUYER DEMAND, REQUIREMENTS CRUD & ROLE OFFERS (FR-4)

Scope: Capture structured buyer procurement requirements (crop, quantity, target price, budget ceiling, delivery window) and provide role-filtered market board views with strict RBAC enforcement.

FEATURE	ENDPOINT	RESULT	RUNTIME EVIDENCE & NOTES
Create Requirement	POST /api/v1/requirements/	PASS	HTTP 200, returned requirement_id: 4c475027-4ba, saved budget ■30,000
List Requirements	GET /api/v1/requirements/	PASS	HTTP 200, returned active requirements list (count: 3)
Get by ID	GET /api/v1/requirements/{id}	PASS	HTTP 200, retrieved requirement payload (crop: Tomato)
Update Target	PATCH /api/v1/requirements/{id}	PASS	HTTP 200, target price updated to ■26.00/kg
Buyer Offer Post	POST /api/v1/buyers/offers	PASS	HTTP 200 with Buyer JWT, creates offer bound to user_id
Role Offer Filter	GET /role-offers/?role=buyer	PASS	HTTP 200, returns buyer-filtered offer array
RBAC Guard (Negative)	POST /buyers/offers (Farmer)	PASS	HTTP 403 Forbidden (detail: Role 'buyer' required)

2. PHASE 4: AI-POWERED MULTI-CRITERIA MATCHING ENGINE (FR-5)

Scope: Evaluate candidate compatibility based on crop matching, price overlap (Zone of Possible Agreement / ZOPA), geographical proximity, and volume thresholds.

FEATURE	ENDPOINT	RESULT	RUNTIME EVIDENCE & NOTES
Listing-to-Buyers	POST /matching/listing-to-buyers	PASS	HTTP 200, total_matches: 2, Top match score 0.85
Requirement-to-Listings	POST /matching/requirement-to-listings	PASS	HTTP 200, evaluated buyer budget ■15,000 and target price ■25.00
Unmatched Crop (Neg)	POST /matching/listing-to-buyers	PASS	HTTP 200, total_matches: 0, matches: [] handled cleanly

3. PHASE 5: AUTONOMOUS MULTI-AGENT NEGOTIATION & LANGGRAPH ENGINE (FR-7, FR-8)

Scope: Orchestrate 9-node LangGraph execution state graph with Ollama LLM (qwen3:8b/qwen2.5:1.5b), iterative concession curves, consensus deal closing, and permanent transaction history logging.

FEATURE	ENDPOINT	RESULT	RUNTIME EVIDENCE & NOTES
LangGraph AI Negotiation	POST /negotiation/start-negotiation	PASS	HTTP 200, status: DEAL, Agreed final price: **■26.25/kg** (ID: neg_cc4633f3)
Telemetry & Status Poll	GET /negotiation-status/{id}	PASS	HTTP 200, returned complete 3-round bid progression and deal summary
Agent Registry	GET /negotiation/agents	PASS	HTTP 200, returned 7 registered agent roles (Farmer, Buyer, Warehouse, etc.)
Ledger History	GET /api/v1/history/{user_id}	PASS	HTTP 200, verified 1 finalized deal persisted in PostgreSQL history table
Invalid Query (Negative)	GET /negotiation-status/invalid_id	PASS	HTTP 404 Not Found (detail: Negotiation not found)

4. PHASE 3 → 5 END-TO-END INTEGRATION CHAIN

A single sequential end-to-end integration journey was executed directly across all newly introduced Phase 3–5 capabilities using fresh live entities:

STEP	FLOW COMPONENT	ROUTE / API	EXPECTED OUTCOME	ACTUAL RUNTIME OUTCOME	STATUS
1	**Farmer Listing**	POST /api/v1/listings/	Listing created with UUID	HTTP 200 (id: 7238f464-e22, Tomato 1200kg @ ■20)	**PASS**
2	**Buyer Requirement**	POST /api/v1/requirements/	Requirement registered	HTTP 200 (id: 4c475027-4ba, Tomato 1000kg @ ■25)	**PASS**
3	**Semantic Matching**	POST /matching/listing-to-buyers	Match score > 0.80	HTTP 200 (total_matches: 2, Score: 0.85)	**PASS**
4	**AI Multi-Agent Negotiation**	POST /negotiation/start-negotiation	LangGraph converges on deal	HTTP 200 (status: DEAL, Agreed: ■26.25/kg)	**PASS**
5	**Telemetry Polling**	GET /negotiation-status/{id}	3-round bid logs returned	HTTP 200 (Telemetry and price series verified)	**PASS**
6	**Ledger Persistence**	GET /history/{user_id}	Deal indexed in user history	HTTP 200 (1 verified record in Postgres history)	**PASS**

5. REAL DEFECTS & OUT-OF-SCOPE ANALYSIS

• **Phase 1–5 Application Bugs: 0 Defects.** All routes in Phases 1 to 5 return HTTP 200 OK with valid schemas.
• **Schema Strictness:** POST /buyers/offers requires buyer_name, min_price, max_price, quantity, location; POST /matching/requirement-to-listings requires budget.
• **RBAC Security Checks:** POST /buyers/offers rejects Farmer tokens with HTTP 403 Forbidden (Working as intended).
• **Out-of-Scope Known Issues:** POST /workflows/plan (Phase 7), GET /transport/estimate (Phase 6), and POST /node/select (Phase 10) are outside Phase 3–5 scope.

6. MAIN APIs FOR MANUAL ACCEPTANCE TESTING

1. Post Buyer Requirement: POST http://localhost:8000/api/v1/requirements/ (Auth: Bearer BuyerToken)
{"crop": "Tomato", "quantity": 1000.0, "target_price": 25.0, "max_price": 28.0, "location": "Mumbai", "budget": 30000.0, "delivery_days": 5, "quality_grade": "A"}
Expected: HTTP 200 with generated requirement_id.

2. Match Listing to Buyers: POST http://localhost:8000/api/v1/matching/listing-to-buyers (Auth: Bearer BuyerToken)
{"crop": "Tomato", "quantity": 1200.0, "min_price": 20.0, "location": "Nashik"}
Expected: HTTP 200 with total_matches >= 1.

3. Start LangGraph Negotiation: POST http://localhost:8000/api/v1/negotiation/start-negotiation (Auth: None)
{"farmer_name": "Ramesh Farmer", "crop": "Tomato", "quantity": 1200, "min_price": 20.0, "shelf_life": 6, "location": "Nashik", "quality": "A", "language": "Marathi"}
Expected: HTTP 200 with status: "DEAL", agreed final_price between ■20 and ■28.

4. Verify Deal in History: GET http://localhost:8000/api/v1/history/{user_id} (Auth: Bearer BuyerToken)
Expected: HTTP 200 with history array containing the finalized negotiation record.

7. CHECKPOINT DECISION & READINESS

STATUS: READY FOR MANUAL PHASE 3–5 TESTING

All Phase 3, Phase 4, and Phase 5 modules, individual routes, negative test cases, and the complete 6-step integration flow have passed with verified runtime evidence.